Positive reduced numerators for Carlitz–Riordan interpolation

Principia Math In Progress

21 September 2026

Task 1: the full numerator assertion

Write $[r] = 1 + q + \cdots + q^{r-1}$, $[0]! = 1$, and $[r]! = \prod_{i=1}^{r}[i]$. Chapoton–Zeng [1] define $C_1 = 1$ and the polynomials $C_n(x \mid q) \in \mathbb{Q}(q)[x]$ by

$$\frac{C_{n+1}(1+qx \mid q) - C_{n+1}(x \mid q)}{1+(q-1)x} = qC_n(q^2x + q + 1 \mid q), \qquad C_{n+1}(-q^{-1} \mid q) = 0.$$

Their Conjecture 8 asks for positive numerators with exact denominator $[n-1]!$. Here positivity means nonnegative coefficients, with every coefficient present in the support positive; it does not require every monomial in a bounding rectangle to occur.

Theorem 1. *For every $n \geq 1$,*

$$P_n(x,q) = [n-1]!\, C_n(x \mid q) \in \mathbb{Z}_{\geq 0}[x,q], \qquad [x^{n-1}]P_n = q^{(n-1)^2}.$$

Every power x^k, $0 \leq k \leq n-1$, has a nonzero coefficient polynomial in q. Moreover P_n and $[n-1]!$ are coprime in $\mathbb{Q}[x,q]$, so this denominator is already reduced.

Prior input. Let $T_0 = 1$ and $T_{r+1} = \sum_{i=0}^{r} q^i T_i T_{r-i}$ be the standard Carlitz–Riordan polynomials, all in $\mathbb{Z}_{\geq 0}[q]$. Theorem 7, equation (4.9), of [1] proves

$$[n]C_{n+1} = ([2n-1] + xq^{2n-1})C_n + \sum_{j=0}^{n-2}[n-j-1]T_j q^{2j+1} C_{n-j}.$$

This recurrence is the authors' result. The proof below is its direct consequence, not a new recurrence.

Proof. Multiply the known recurrence by $[n-1]!$. It becomes

$$\begin{aligned} P_{n+1} = {} & ([2n-1] + xq^{2n-1})P_n \\ & + \sum_{j=0}^{n-2} q^{2j+1}[n-j-1]T_j \left(\prod_{t=n-j}^{n-1} [t] \right) P_{n-j}. \end{aligned}$$

Every multiplier is a polynomial with nonnegative integer coefficients. Empty products equal one and the sum is empty for $n = 1$. Induction from $P_1 = 1$ proves coefficient positivity for all n. The first term preserves every existing x-degree and adds the next one, proving nonvanishing in each degree.

To finish the proof, all summands except $xq^{2n-1}P_n$ have x-degree at most $n-1$. Thus the leading coefficient obeys $L_{n+1} = q^{2n-1}L_n$, $L_1 = 1$, yielding $L_n = q^{1+3+\cdots+(2n-3)} = q^{(n-1)^2}$. Any common irreducible factor of P_n and $[n-1]!$ must belong to $\mathbb{Q}[q]$ and divide every x-coefficient of P_n, including this monomial. But $[n-1]!(0) = 1$, so it has no factor q. No common nonconstant factor exists. This also covers $n = 1$. □

For example,

$$P_2 = 1 + qx, \qquad P_3 = (1+q)^2 + (q + 2q^2 + 2q^3)x + q^4x^2.$$

The argument establishes both parts of the conjecture for the entire sequence, including irreducibility of the fraction; it does not infer the infinite assertion from computed examples.

Attribution and campaign status

The complete current 15-page preprint [1] and the complete 14-page author note [2] were read. Cigler's Theorem 5 gives a positive expansion in q-Gould polynomials of parameter one, while equation (3.1) identifies the present family. This is an additional substantial prior implication that must be credited in any priority assessment. The source's own recurrence already supplies the short proof above.

The identified 14-page journal final, *Contributions to Discrete Mathematics* 10(1) (2015), 99–112, could not be read: public download requests returned server errors. Accordingly this package receives **zero new campaign credit**, pending final-source and prior-implication review. No new research priority is claimed. The exact verifier checks the original defining difference equations, the boundary condition, coefficient signs, leading terms, and denominator coprimality for initial degrees. These checks supplement the proof. Review is by the authoring agent only.

Proposed follow-up Tasks 2–5

1. **Task 2.** Construct a direct combinatorial model for the coefficients of the reduced numerator.
2. **Task 3.** Determine the exact monomial support and Newton polygon for every n.
3. **Task 4.** Classify unimodality or log-concavity of the coefficient polynomials in q for fixed powers of x.
4. **Task 5.** Extend the exact reduced-denominator statement to Cigler's general parameter-m Gould family.

References

[1] F. Chapoton and J. Zeng, *A curious polynomial interpolation of Carlitz–Riordan's q-ballot numbers*, arXiv:1311.7228v2 (2013). Journal final: Contrib. Discrete Math. 10(1) (2015), 99–112, unread.

[2] J. Cigler, *Some notes on q-Gould polynomials* (2013), author PDF.